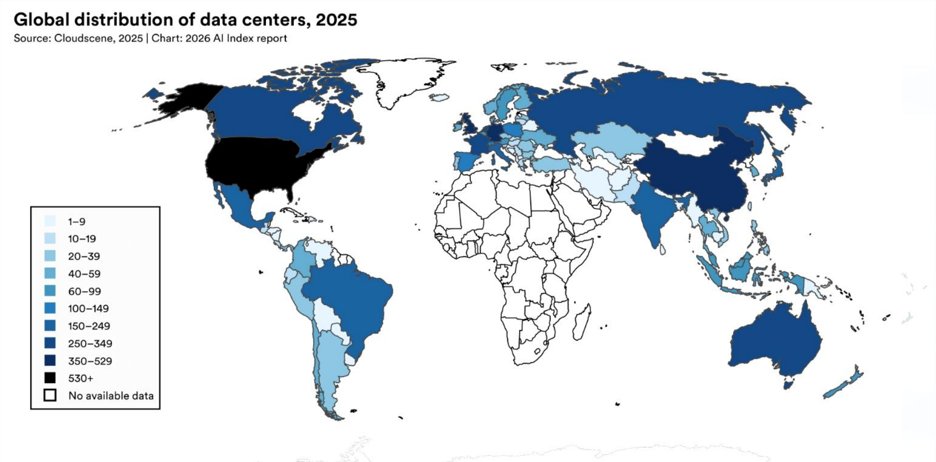
Execute um modelo de IA Generativa localmente

usando Python e Ollama

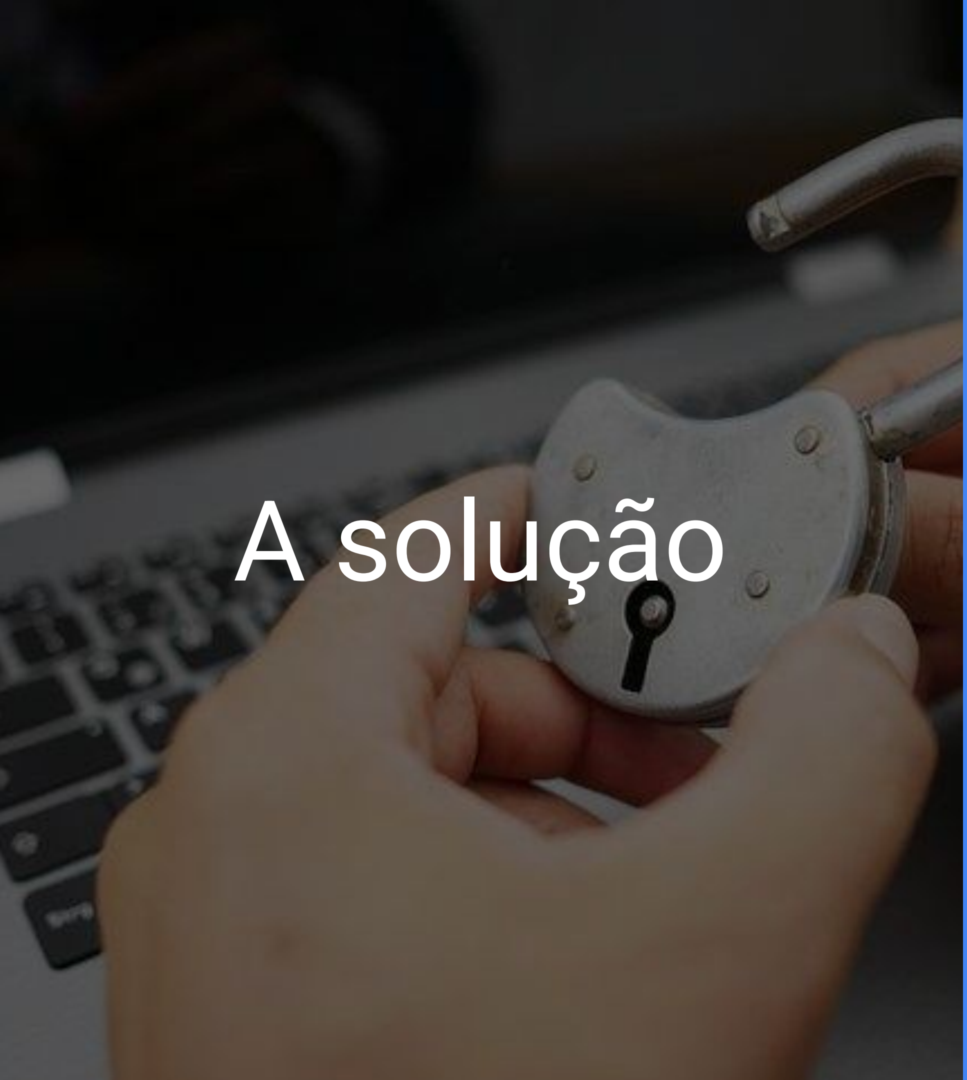

⚠️Alerta de tópico sensível

O problema

1. dados usados com permissão?
2. quem gera dados tem remuneração?
3. a privacidade na rede é preservada?
4. o que aconteceu com *whistleblower*?
5. é seguro para a saúde mental?
6. ajuda no desenvolvimento intelectual?
7. faz bem para o meio ambiente?
8. faz sentido substituir pessoas?



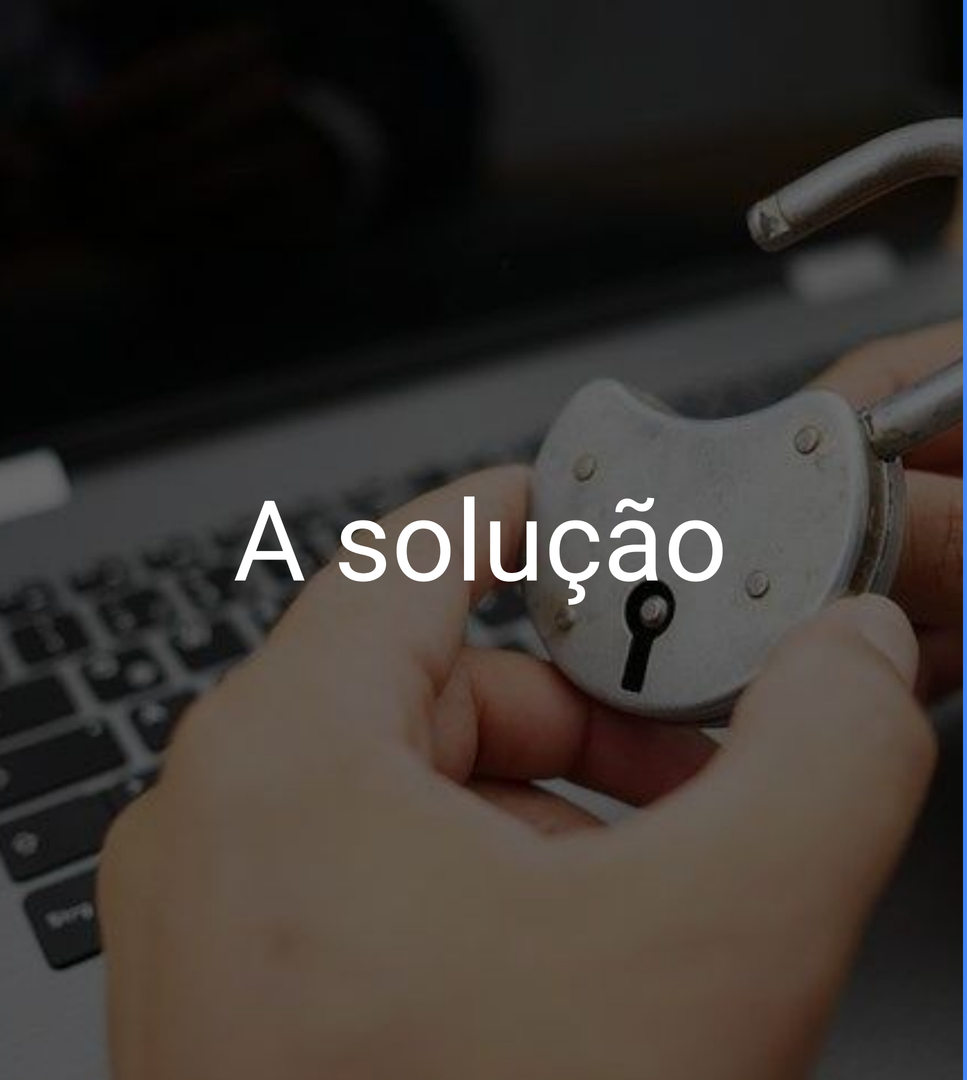
https://hai.stanford.edu/ai-index/2026-ai-index-report



A solução

Evitar o uso
caso não tenha jeito,
pelo menos proteger a privacidade e um pouco do seu mental executando modelos de IA Generativa localmente

A solução



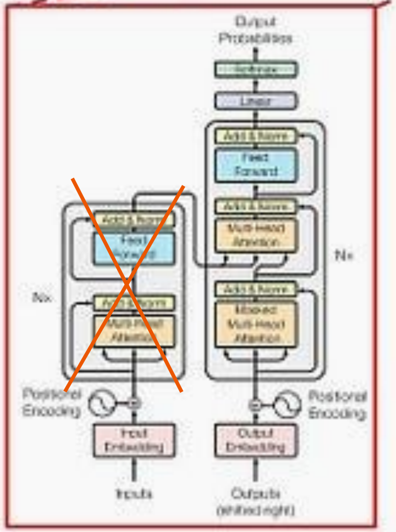
Neste caso,
seria ideal poder treinar modelos localmente,
só que para isso é necessário GPU da NVIDIA ou chips MAC M1/M2/M3/M4
e muita memória disponível

Tipos de treinamentos

post: https://www.sundeepteki.org/advice/the-complete-guide-to-post-training-llms-how-sft-rlhf-dpo-and-grpo-shape-llms

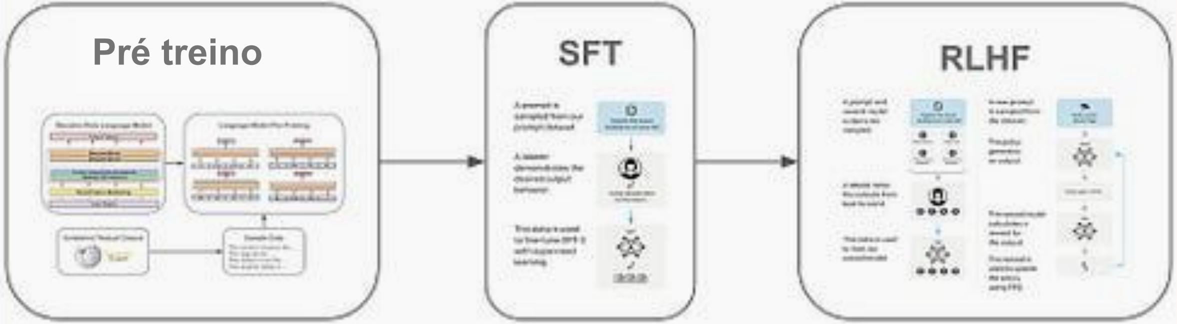
video: https://www.youtube.com/watch?v=IrBhHPmv0TQ&t=446s

IA Generativa = Transformers + Pré-treino + Alinhamento/ pós-treino



Apenas decodificador (decoder only)

Treinamento de LLM



feito com transformadores

Ajuste Supervisionado

Supervised Fine Tuning

Aprendizagem de reforço a partir de feedback humano

Reinforcement Learning from Human Feedback

https://www.linkedin.com/pulse/how-does-chatgpt-work-under-hood-marco-van-hurne-vl6ve/

Como treinar “local”

Unsloth

Notebooks para usar com Colab:
https://unsloth.ai/docs/get-started/unsloth-notebooks

Exemplos no TRL do HuggingFace:
https://huggingface.co/docs/trl/index

🛠️ Unsloth Requirements

Here are Unsloth's requirements including system and GPU VRAM requirements.

Unsloth can be used in two ways: through Unsloth Studio, the web UI, or through Unsloth Core, the original code-based version. Each has different requirements.

Unsloth Studio Requirements

- **Mac:** Training, MLX and GGUF inference are ALL supported.
- **CPU: Unsloth still works without a GPU**, for Chat + Data Recipes.
- **Training:** Works on **NVIDIA**, **AMD**, **Intel** GPUs and **Mac** devices

Windows

Unsloth Studio works directly on Windows without WSL. To train models, make sure your system satisfies these requirements:

Requirements

- Windows 10 or Windows 11 (64-bit)
- NVIDIA GPU with drivers installed
- **App Installer** (includes `winget`): here ↗
- **Git:** `winget install --id Git.Git -e --source winget`

só não é pré-requisito para mac

Como treinar “local”

Unsloth

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https://huggingface.co/docs/trl/index

Taxonomy

Below is the current list of TRL trainers, organized by method type (⚡ = vLLM support; 🧪 = experimental).

Online methods

- GRPOTrainer ⚡
- RLOOTrainer ⚡
- OnlineDPOTrainer 🧪 ⚡
- NashMDTrainer 🧪 ⚡
- PPOTrainer 🧪
- XPOTrainer 🧪 ⚡

Offline methods

- SFTTrainer
- DPOTrainer
- KTOTrainer
- BCOTrainer 🧪
- CPOTrainer 🧪
- ORPOTrainer 🧪

Reward modeling

- RewardTrainer
- PRMTrainer 🧪

Knowledge distillation

- DistillationTrainer ⚡
- GKDTrainer 🧪
- MiniLLMTrainer 🧪

You can also explore TRL-related models, datasets, and demos in the TRL Hugging Face organization.

github.com/ollama/ollama-python/blob/main/examples/chat-with-history.py

Files

main

Go to file

- .github
- examples
  - README.md
  - async-chat.py
  - async-generate.py
  - async-structured-outputs.py
  - async-tools.py
  - chat-logprobs.py
  - chat-stream.py
  - chat-with-history.py
  - chat.py
  - create.py
  - embed.py
  - fill-in-middle.py

Ollama

Code Blame 35 lines (32 loc) · 1.21 KB

```python
from ollama import chat

messages = [
  {
    'role': 'user',
    'content': 'Why is the sky blue?',
  },
  {
    'role': 'assistant',
    'content': "The sky is blue because of the way the Earth's atmosphere scatters sunlight.",
  },
  {
    'role': 'user',
    'content': 'What is the weather in Tokyo?',
  },
  {
    'role': 'assistant',
    'content': """The weather in Tokyo is typically warm and humid during the summer months, with tem
    rarely dropping below freezing. The city is known for its high-tech and vibrant culture, with man
  },
]

while True:
  user_input = input('Chat with history: ')
  response = chat(
    'gemma3',
    messages=[*messages, {'role': 'user', 'content': user_input}],
```

Passo a passo

1. Instalar o Ollama

Acesse https://ollama.com/download e escolha o seu sistema operacional

2. Verificar versão do Ollama

No terminal (Mac/Linux) ou no powershell (Windows), digite ollama --version para verificar qual versão foi instalada

3. Baixar e rodar o modelo

Digite ollama run smollm e baixe o modelo. Sempre veja as suas configurações em https://ollama.com/search para ver se tem espaço para o modelo

4. Usar o modelo

Pode realizar as perguntas livremente. O Ollama salva um arquivo de histórico chamado *history* com todas as interações

1. Instalar o Ollama

```
cassia@galadriel:~$ curl -fsSL https://ollama.com/install.sh | sh
>>> Installing ollama to /usr/local
>>> Downloading ollama-linux-amd64.tar.zst
######################################################################## 100.0%
>>> Creating ollama user...
>>> Adding ollama user to render group...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
>>> Enabling and starting ollama service...
Created symlink /etc/systemd/system/default.target.wants/ollama.service → /etc/s
ystemd/system/ollama.service.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
WARNING: No NVIDIA/AMD GPU detected. Ollama will run in CPU-only mode.
```

2. Verificar versão do Ollama

```
cassia@galadriel:~$ ollama --version
ollama version is 0.32.15
cassia@galadriel:~$
```

3. Baixar e rodar o modelo

```
cassia@galadriel:~$ ollama run smollm
pulling manifest
pulling 6cafb858555d: 100%                    990 MB/990 MB   48 MB/s      0s
pulling 1c3c04e4a658: 100%                     485 B
verifying sha256 digest
writing manifest
success
>>>
```

4. Usar o modelo

```
cassia@galadriel:~$ ollama run smollm
>>> hello
Hello! How can I help you today?

>>> are there negative consequences of generative ai use?
Yes, there are several negative consequences of generative AI use,
including job displacement, bias in decision-making, and the potential for
misuse.

>>> Send a message (/? for help)
```

Prática no Colab

1. Login na conta google
2. Abrir o google colab
3. Criar um notebook
4. Seguir o passo a passo

materiais:
https://github.com/cassiasamp/each_ssi_2026

Muito obrigada

Contatos

in @cassiasamp

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